

ARCHAEOLOGY



HUMANKIND HAS ALWAYS been fascinated by the question of who we are, where we came from, and how we used to live. Archaeology is the study of our past, from early prehistory onward, using the material remains of our ancestors and the possessions they left behind. Over thousands of years, evidence of human activity, such as camp fires, rubbish tips, and dwellings, become buried. Archaeological teams discover these sites and uncover this evidence by careful excavation. The material is then conserved and studied in order to help the archaeologist piece together a picture of how people lived and died in the past.

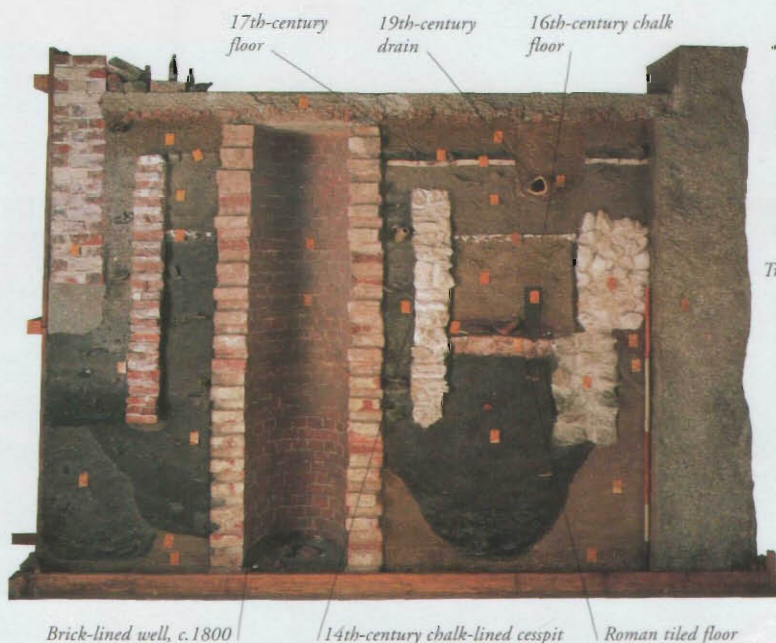
Excavation

Archaeological sites are excavated by layers. Workers remove the top, most recent layer and work down, uncovering older, deeper levels. The study of these layers and the items they contain is called stratigraphy.

Stratigraphy

By revealing features such as ditches, post holes, and floors, stratigraphy gives information about the history of a site, and the people who lived there. In urban areas, such as London, surface levels rise as debris is shovelled in to level the ground before rebuilding. Because it shows a chronological sequence, stratigraphy was used to date sites before radiocarbon dating was invented.

Cross section through a dig, City of London



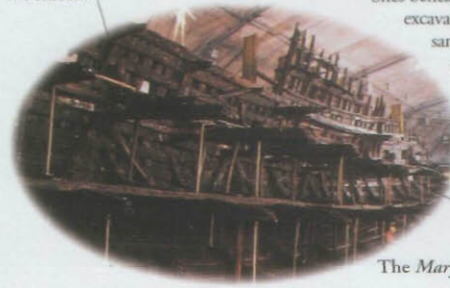
Salt water has caused corrosion

Pewter jug

Investigation

Buried objects are fragile, and decay quickly after excavation. To stabilize them, they are cleaned and conserved. After conservation, an object can be studied. The material of which it is made, its function, and its date are recorded. It may then be photographed and displayed in a museum.

A cradle hoisted the ship from the seabed.



Underwater archaeology

Sites beneath the sea or in lakes are more difficult to excavate than those on land because shifting silt or sand causes poor visibility. However, marine sites often preserve materials, such as the wood of the 16th-century ship, the *Mary Rose*, which would usually be lost on dry land. Conservation may involve treatment with water, sealing with chemicals, or careful drying.

To conserve the wood, chilled water is sprayed on the ship 20 hours a day.

The *Mary Rose* in dry dock

Timeline

1748 Pompeii discovered.

1799 An officer in Napoleon's army discovers the Rosetta Stone, which features 6th-century BC hieroglyphs.

1812 Abu Simbel discovered.

1822 Scholars decipher Egyptian hieroglyphs.

1861 Evans and Prestwich confirm the antiquity of humans, and humans' association with extinct animals.

1891 *Homo erectus* material found.

1922 Howard Carter discovers the tomb of Tutankhamun.

1931 Louis Leakey begins excavations at Olduvai Gorge.

1940 Archaeologists discover prehistoric Lascaux cave paintings.

1949 Radiocarbon dating is developed.

1974 Donald Johanson discovers "Lucy", an early hominid.



Australopithecus, an early human ancestor

Mortimer Wheeler

The greatest field archaeologist of the day, Wheeler (1890–1976) set up the Institute of Archaeology, London. He developed new excavation methods, and made archaeology popular through TV. In 1944, he became Director-General of Archaeology in India, and investigated the Indus Valley Civilization.



Discovery

Archaeological sites are found during building work, through reading historical documents, geophysical surveys (the study of the soil's structure), and field walking (recording above-ground objects).



Iron Age fort, England

Aerial photography

Horizontal and vertical lines seen from the air often show medieval strip fields, ancient roads, walls, and ditches. Aerial photography done when the sun is low shows varying surface levels, moisture levels, and vegetation most clearly.



Trowel

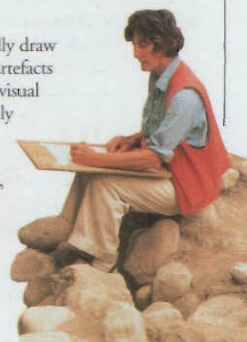
Measuring tape

Tools

Archaeologists use shovels and handpicks to remove the topsoil. Then smaller hand tools are used, such as dental picks, teaspoons, and trowels, to excavate delicate objects.

Finds

Archaeologists usually draw or photograph the artefacts (objects) to make a visual record. They carefully measure and record the shapes, colours, decorations, and ages of any artefacts or features. This helps archaeologists link and relate different objects and sites.



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